

28.

**Ans: B**

$K_{\max} = 0$  eV occurs when  $\lambda = 300$  nm. Therefore, the threshold wavelength is 300 nm.

By photoelectric equation,

$$\begin{aligned}\Phi &= \frac{hc}{\lambda} - E_{k\max} \\ &= \frac{6.63 \times 10^{-34} (3.0 \times 10^8)}{300 \times 10^{-9}} - 0\end{aligned}$$

$$\begin{aligned}\Phi &= 6.63 \times 10^{-19} \text{ J} \\ &= 4.14 \text{ eV}\end{aligned}$$



